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RESEARCH ARTICLE

Adaptive Circular Receptive Field for Remote Sensing Image Target Detection

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ABSTRACT Remote sensing images contain a variety of targets. The circular and square-like targets are very common. However, general methods are prone to incorporating background information and have problems with rotation angles when detecting circular and square-like targets. The Adaptive Circular Receptive Field Network (ACRFNet) is proposed to address these problems. Firstly, a radius detection head based on the center point is designed, which predicts the center point and radius of the target to obtain a bounding circle. Compared with the rectangular bounding box, the bounding circle contains less background information and has rotation invariance, achieving in more accurate localization. Secondly, in order to better extract the features of circular targets and cope with scale changes, a multi-scale feature fusion method based on adaptive circular convolution is proposed. Adaptive circular receptive fields replace the ordinary convolution of square receptive fields, and a feature pyramid structure is introduced to enrich the semantic information of the feature map. Finally, in order to reasonably evaluate the performance of the model, Circle IOU is used to calculate mAP to fit the bounding circle. The experimental results on the NWPU VHR-10 and DIOR datasets demonstrate that the model has significantly improved detection accuracy for circular and square-like targets and has better generalization for other types of targets.

INDEX TERMS Remote sensing image, target detection, bounding circle, adaptive circular receptive field.

I. INTRODUCTION

In recent years, satellite remote sensing technology has developed rapidly. Remote sensing image detection has been widely applied in various fields, playing an important role in geological exploration, resource exploration, environmental monitoring, military reconnaissance, and other areas [1]. Although the acquisition technology of remote sensing images has gradually matured, the interpretation technology of remote sensing images has not developed at a similar pace. Remote sensing images have the characteristics of a large imaging area, complex background information, and less available information. Furthermore, remote sensing images contain a variety of targets, increasing the demand

for designing specific detectors for specific targets in remote sensing images. Many industrial structures appear as circular and square-like shapes in remote sensing images, such as storage tanks and nuclear wastewater storage tanks. Therefore, it is important to develop detectors for circular and square-like targets in remote sensing images to precisely detect industrial pollution structures for industrial planning and environmental protection.

Within the realm of object detection, there exist two predominant methodologies: traditional methods and deep learning. Today, deep learning has emerged as a principal approach in the field of object detection. The first work using deep learning in the field of object detection was R-CNN proposed by Girshick et al. [2]. The R-CNN extracts a large number of region proposals in the image initially and then uses a convolutional neural network to

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extract features for each region proposal. These features are classified by the classifier subsequently. Based on R-CNN, Girshick [3] proposed Fast R-CNN, and Ren et al. [4] proposed a more efficient two-stage object detection algorithm, Faster R-CNN. In pursuit of faster detection speeds, Redmon et al. [5], [6] proposed the YOLO series of algorithms, and Lin et al. [7] proposed RetinaNet. These are typical one-stage object detection algorithms. One-stage object detection algorithms eliminate the step of generating candidate regions, and classification and localization are conducted in the same stage. Compared to two-stage object detection algorithms, one-stage object detection algorithms are faster, require fewer computational parameters, and can also achieve better detection accuracy. To further simplify the network structure, Zhou et al. [8] proposed CenterNet, an anchor-free object detection algorithm. In contrast to the anchor-based algorithms, CenterNet does not utilize prior anchors. Instead, it directly predicts the center points, width, and height of the objects, which are then used to regress the bounding boxes. Therefore, CenterNet has a more concise network structure, faster detection speed, and achieves better detection performance. Other common anchor-free algorithms include FCOS proposed by Tian et al. [9], ExtremeNet proposed by Zhou et al. [10], and Reppoints proposed by Yang et al. [11].

Although the aforementioned object detection algorithms have demonstrated impressive detection performance in general targets, general object detectors still face challenges in dealing with the complex backgrounds and large-scale variations in remote sensing images. Multi-scale feature fusion and different receptive field mechanisms are effective ways to improve the performance of object detection in remote sensing images and other downstream tasks of remote sensing images [12], [13]. Zhou et al. [14] address the problems of dense arrangement of small targets and complex backgrounds in remote sensing images. Firstly, the Feature Reuse Module (FRM) is proposed to integrate the semantic and location information in the feature map. This can improve the ability of the model to detect small targets. Secondly, DFF-PANet is designed to enhance the ability of the model to fuse feature maps of different scales. For the small target of manhole covers in remote sensing images, Liu et al. [15] redesigned the feature extractor based on the VGG network to increase the variety of receptive field size. Then two sub-networks, a multi-scale output network (MON) and a multi-level convolution matching network (M-CMN), are used to fuse the feature maps of different scales. In this way, the model can detect small targets accurately.

General remote sensing image target detection methods can detect most targets, but remote sensing images contain a wide variety of targets, each with different characteristics. Therefore, researchers have proposed numerous works targeting specific targets in remote sensing images. Liu et al. [16] integrated a new feature extractor based on YOLOv5 for remote sensing image aircraft target detection. The

coordinate attention was incorporated into the network. Additionally, a combination of hybrid dilated convolution and a redesigned residual structure was used to improve feature extraction at shallow levels and the retrieval of location information. The feature expression capability of the network for different scale targets is optimized. In response to the challenges of small target scale and complex backgrounds in ship target detection from remote sensing images, Huang et al. [17] introduced dilated convolutions based on YOLOv4 to expand the receptive field and used attention mechanisms to adjust the weights of different features. It improved the ability of the model to detect small targets and handle complex backgrounds.

Enhancing the feature extraction capabilities of the network by incorporating target characteristics is the mainstream approach in specific target detection tasks for remote sensing images. It is also a common direction for improvement in other target detection tasks. Yang et al. [18] proposed CircleNet for glomerulus detection, which converts a rectangular bounding box into a bounding circle. This approach better fitted the glomerulus targets, suppressed background information interference, and achieved a higher degree of accuracy in classification and localization. Zheng et al. [19] used a polygon detector for various targets such as crosswalks and traffic signs on autonomous driving roads. The model achieved high detection accuracy and met the real-time requirements needed for autonomous driving. Qi et al. [20] proposed dynamic snake convolution for remote sensing roads and cardiovascular structures with slender and irregular orientations. The snake receptive field is achieved to extract the features of tubular targets more effectively by altering the direction and size of the deformable convolution.

In the field of remote sensing image target detection, horizontal rectangular bounding boxes contain more background information when detecting circular or square-like targets such as tanks or nuclear waste storage tanks. Additionally, horizontal rectangular bounding boxes can be influenced by the rotation angle. The CRNet we previously proposed has confirmed that the bounding circle can effectively address these problems, but there are still problems with large-scale variations as well as dense alignments [21]. Therefore, in order to better detect circular and square-like targets in remote sensing images, the Adaptive Circular Receptive Field Remote Sensing Image Target Detection method (ACRFNet) is proposed, which includes the following three main contributions:

- (1) For circular and square-like targets, the bounding circle is proposed to reduce background information interference. This also avoids problems with target rotation angles.

- (2) An adaptive circular receptive field convolution is proposed, which converts the receptive field of convolution into a circular shape and achieves adaptive expansion or contraction. Combining it with a feature pyramid enables the network to have superior feature representations of circular and square-like targets. Moreover, the

ability of the network to handle scale variations is also enhanced.

(3) Circle-IOU is proposed in order to calculate the IOU of the bounding circle and better evaluate model performance.

II. METHODS

A. FRAMEWORK OF ACRFNET

The adaptive circular receptive field for the remote sensing circular target detection network is proposed based on the anchor-free target detection network CenterNet. The overall framework is shown in Fig. 1. First, the convolutional kernel with Adaptive Circular Receptive Field is designed to better extract features of circular targets. A feature pyramid is introduced into the network structure. Applying the convolutional kernel with the adaptive circular receptive field to the deep lateral connections of the pyramid can integrate deep abstract semantic information into the shallow layers through the pyramid structure, enriching the semantic information contained in the feature map. Secondly, a circular radius detection head is proposed to replace the width-height detection head in order to transform the rectangular bounding box into the bounding circle. The output of the detection head is fed to the decoder. The bounding circle is obtained to detect targets after decoding. Finally, Circle-IOU is used to calculate the intersection over union (IOU) between two circular boxes to reasonably evaluate the performance of the model.

B. BOUNDING CIRCLE LABEL

The transformation of a traditional bounding box into a bounding circle not only requires the design of a radius detection head but also the calculation of the intersection over union (IOU) between two circular boxes. Therefore, the rectangular bounding box labels need to be converted into bounding circle labels. As shown in Fig. 2. The label format of the rectangular bounding box is composed of the top-left corner and the bottom-right corner. The bounding circle labels consist of a center point and a radius. However, directly modifying the horizontal bounding box labels not only involves a significant amount of work but also easily leads to errors due to scaling operations. Therefore, after completing the scaling operation, the scaled rectangular bounding box labels are used to calculate the center coordinates and radius, rather than directly modifying the labels of the rectangular bounding box. Its calculation formula is as follows:

$$x = \frac{x_2 - x_1}{2}, y = \frac{y_2 - y_1}{2}, \quad (1)$$

$$r = \frac{\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}}{2} \quad (2)$$

where x and y are the coordinates of the center point. And r represents the radius of the circular box.

C. ADAPTIVE CIRCULAR RECEPTIVE FIELD

Inspired by circular convolution [22], we propose a convolution with an adaptive circular receptive field to reduce

the interference of background information and improve the ability of the model to extract features of circular and square-like targets. As shown in Fig. 3. First, the square receptive field in Fig. 4 is converted into a circular receptive field. Then, the concept of deformable convolution is introduced to predict the offset of the circular edge position, achieving the adaptive expansion or contraction of the circular receptive field. In Fig. 3, the solid circular receptive field can adaptively expand to the outermost dashed large circle or contract to the innermost dashed small circle. This can better fit the target, enabling better extraction of its internal features and adapting to scale changes.

The specific steps for the adaptive circular receptive field are as follows: First, the traditional 3×3 convolutional receptive field is square. Its calculation formula is as follows:

$$y(p_0) = \sum_{p_n \in R} w(p_n) x(p_0 + p_n). \quad (3)$$

where p_0 represents the center of the convolution kernel. p_n is the coordinate of each point in the convolution grid relative to the center of the convolution kernel. $w(p_n)$ is the weight coefficient. The coordinate set of R is as follows:

$$R = \left\{ \begin{array}{l} (-1, 1), (0, 1), (1, 1), (-1, 0), (0, 0), \\ (1, 0), (-1, -1), (0, -1), (-1, -1) \end{array} \right\}. \quad (4)$$

Then, offset the midpoint coordinates of the four edges outward to the circumcircle position, as shown in Fig. 4.

The receptive field shape of the convolutional kernel changes from square to circular. The set of coordinates after offset is as follows:

$$C = \left\{ \begin{array}{l} (-1, 1), (0, \sqrt{2}), (1, 1), (-\sqrt{2}, 0), (0, 0), \\ (\sqrt{2}, 0), (-1, -1), (0, -\sqrt{2}), (-1, -1) \end{array} \right\}. \quad (5)$$

Furthermore, using the idea of deformable convolution, the offset Δc_n is added to the circular convolution kernel. Changing the eight coordinates other than the center position based on the predicted offsets. This allows the receptive field of the circular convolution kernel to achieve adaptive scaling and focus on the target area to better extract target features. The adaptive circular receptive field is as follows:

$$y(c_0) = \sum_{c_n \in C} w(c_n) x(c_0 + c_n + \Delta c_n) \quad (6)$$

where C is the coordinate set of each sampling position in the circular convolution kernel. $w(c_n)$ is the weight coefficient of each position of the convolution kernel. c_0 is the sampling position of the convolution kernel center in the feature map. c_n is the coordinate of each position relative to the center of the convolution kernel in the convolution grid. Δc_n is the offset for each position in the adaptive circular convolution kernel. $c_0 + c_n$ represents the conversion of a square receptive field to a circular receptive field. The offset Δc_n is added to each position on the circular edge to achieve an adaptive change in the circular receptive field.

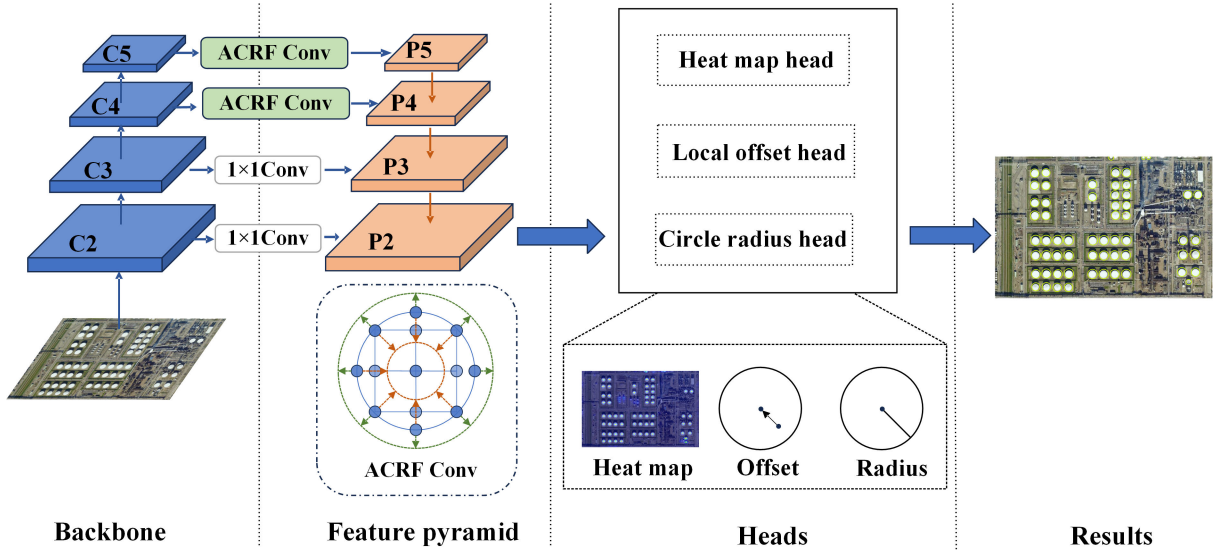


FIGURE 1. The framework of ACRFNet.

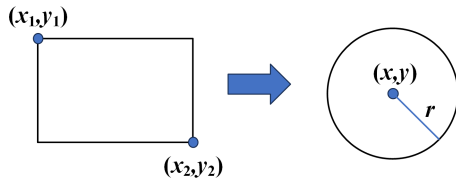


FIGURE 2. Conversion of rectangular label data to circular label data.

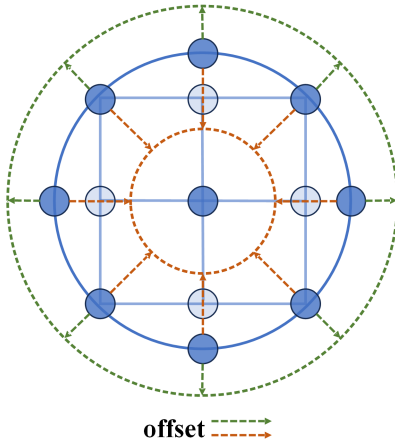


FIGURE 3. Adaptive circular receptive field.

The coordinates after the offset have decimal values, but the pixel positions in the feature map are integers. It is not possible to use pixel points to represent the position after the offset. Therefore, bilinear interpolation is adopted to calculate the weighted sum of the four pixel values around the offset position. The position after the offset is represented by the pixel-weighted sum. Its calculation formula is as

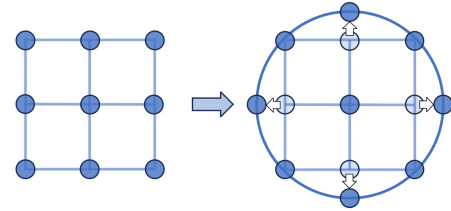


FIGURE 4. Conversion from rectangular receptive field to circular receptive field.

follows:

$$x(p) = \sum_q \frac{\max(0, 1 - |q_i - p_i|)}{\max(0, 1 - |q_j - p_j|)} x(q) \quad (7)$$

where $x(p)$ is the pixel value at the interpolation point p . $x(q)$ is the neighborhood pixel value of the interpolation point p . The $\max()$ function indicates that the distance from the neighboring pixel point q to the interpolated pixel point p does not exceed 1. And the distance is used as the weight of the neighborhood pixel value $x(q)$. The pixel value at the interpolation point is the weighted sum of the four neighboring pixel values.

D. FEATURE PYRAMID STRUCTURE

The original backbone network structure of CenterNet is shown in Fig. 5. The backbone network extracts the input image features to obtain a feature map C5 of size 16×16 . Since CenterNet uses heatmaps for classification and center point selection, feature maps with an excessively small scale often result in peak points overlapping within the heatmap. This affects the quality of center point selection. Therefore, the feature map is upsampled three times to obtain a feature

map of size 128×128 , which is then input to the detection head for classification and regression.

The network structure with the added feature pyramid is shown in Fig. 5b. The three simple upsampling operations of CenterNet have been replaced by the feature pyramid. First, the feature maps at layers C2 to C5 are extracted from the backbone. Lateral connection operations are used to unify the number of channels in the feature maps. Then, the adaptive circular receptive field is applied to the lateral connections between the deep C5 layer and the C4 layer. Applying adaptive circular receptive fields at deeper levels enables the convolution kernel to extract more appropriate circular target features without increasing too much computational load. Finally, the P5 layer is fused with the C4 layer to obtain the P4 layer through upsampling and feature fusion operations. After a total of three upsampling and feature fusion operations, the P2 layer is obtained. In this way, the deep abstract semantic information is fused to the shallow layer, which enriches the feature map information that will be input into the detection head.

E. LOSS FUNCTION

The loss function includes classification loss, radius loss, and center point offset loss. The classification loss L_{cls} is formulated as focal loss to alleviate the class imbalance problem:

$$L_{cls} = \frac{-1}{N} \sum_{xys} \left\{ \begin{array}{l} (1 - \hat{Y}_{xys})^\alpha \ln(\hat{Y}_{xys}) \\ (1 - Y_{xys})^\beta (\hat{Y}_{xys})^\alpha \ln(1 - \hat{Y}_{xys}) \end{array} \right. \quad (8)$$

where N represents the number of targets. Y_{xys} is the ground truth, and $\hat{Y}_{xys}$ is the heatmap used for classification. The xy is the coordinates of the center point. s denotes the quantity of channels within the heatmap, where each channel corresponds to a category in the dataset. α and β are hyperparameters. Equation (8) takes the upper half when $\hat{Y}_{xys} = 1$. Otherwise, Eq.(8) takes the lower half.

The radius detection head is designed for radius prediction, and the radius loss function is as follows:

$$L_{radius} = \frac{1}{N} \sum_{k=1}^N |\hat{S}_{pk} - s_k| \quad (9)$$

where $\hat{S}_{pk}$ is the predicted radius. s_k is the radius of the ground truth. Based on the predicted radius and the radius of the ground truth, the radius loss can be obtained by calculating the difference.

During the upsampling process, the center point coordinates are generally in non-integer form. After each upsampling operation, there is a rounding operation, which causes a deviation in the center point coordinates. Therefore, the center point offset is used to correct the center point coordinates, and the center point offset loss is as follows:

$$L_{offset} = \frac{1}{N} \sum_p \left| \hat{O}_{\tilde{p}} - \left(\frac{p}{M} - \tilde{p} \right) \right| \quad (10)$$

where p is the center point of the bounding circle. M is the upsampling factor. $\tilde{p}$ is the result of flooring, and $\hat{O}_{\tilde{p}}$ is the predicted offset.

The radius loss and offset loss are added together to form the localization loss. The classification loss and localization loss are weighted and added together to form the total loss. The calculation of the total loss is as follows:

$$L_{det} = L_{cls} + \lambda_{size} L_{radius} + \lambda_{offset} L_{offset} \quad (11)$$

F. CIRCLE-IOU

Mean Average Precision (mAP) is a common metric for evaluating models. Before calculating mAP, the IOU between the predicted box and the ground truth needs to be computed first. As the rectangular bounding box is transformed into the bounding circle, the calculation of the IoU between two rectangular boxes is transformed into the calculation of the IoU between two circular boxes. Therefore, Circle-IOU is designed to calculate the intersection and union areas of two circular boxes. The Circle-IOU is as follows:

$$\text{Circle-IOU} = \frac{\text{Area}(A \cap B)}{\text{Area}(A \cup B)} \quad (12)$$

where A and B represent two circular boxes. A represents the predicted box and B is the ground truth. As shown in Fig. 6, the center of circle A is at coordinates (A_x, A_y) with a radius r_A , and the center of circle B is at (B_x, B_y) with a radius r_B .

The distance between the two centers is d , which can be calculated using the coordinates of the two centers. The calculation formula is as follows:

$$d = \sqrt{(B_x - A_x)^2 + (B_y - A_y)^2} \quad (13)$$

The perpendicular distance from the intersection points of the two circles to the line connecting the two circle centers is L_y , and the distance from the center of circle A to the perpendicular line is L_x . Its calculation formula is as follows:

$$L_x = \frac{r_A^2 - r_B^2 + d^2}{2d}, L_y = \sqrt{r_A^2 - L_x^2} \quad (14)$$

The area of the union of two circles can be calculated as follows:

$$\text{Area}(A \cup B) = \pi r_A^2 + \pi r_B^2 - \text{Area}(A \cap B) \quad (15)$$

The area of the intersection of two circles can be calculated as follows:

$$\begin{aligned} \text{Area}(A \cap B) = & r_A^2 \sin^{-1}\left(\frac{L_y}{r_A}\right) + r_B^2 \sin^{-1}\left(\frac{L_y}{r_B}\right) \\ & - L_y(L_x + \sqrt{r_A^2 - r_B^2 + L_x^2}) \end{aligned} \quad (16)$$

III. EXPERIMENT AND ANALYSIS

A. DATASETS

To verify the effectiveness of ACRFNet, the NWPU VHR-10 [23] and the DIOR datasets [24] are used for the experiment. The NWPU VHR-10 contains 800 images. Among them, 650 images contain annotation information, and 150 images do not. There are a total of 3,775 object instances in

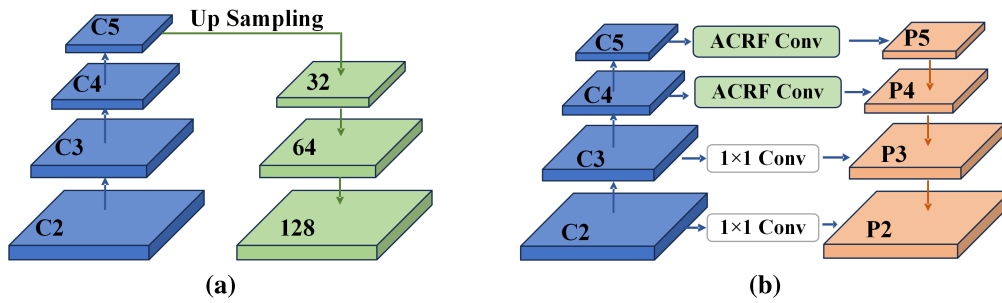


FIGURE 5. Comparison of backbone network structures:(a) CenterNet; (b) ACRFNet.

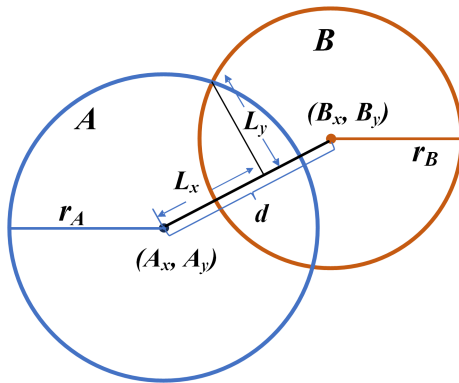


FIGURE 6. Circular intersection and union area.

10 categories, including airplane, baseball fields, storage tank, basketball court, ground track field, bridge, vehicle, and tennis court. The DIOR dataset is a large-scale optical remote sensing dataset for target detection in remote sensing images. This dataset contains 23,463 images, 19,242 object instances, and a total of 20 categories. These categories include airport, chimney, dam, expressway service area, expressway toll station, golf course, overpass, train station, windmill, stadium, and more.

B. EXPERIMENTAL ENVIRONMENT AND EVALUATION INDICATORS

The experiment is conducted in a Windows 10 system environment. The CPU is Intel Xeon(R) Gold 5128R@2.10GHz. The graphics card is an Nvidia GeForce RTX 3090 with 24GB of VRAM. The Python version is 3.7. The PyTorch version is 1.7.1. The CUDA version is 11.2. The NWPU VHR-10 dataset and the DIOR dataset are each divided into training, validation, and test sets, with a ratio of 7:2:1.

Average Precision (AP) and mean Average Precision (mAP) are used to evaluate the accuracy of the model. Average Precision is used to evaluate the detection accuracy of the model for a single category. The mean average precision is used to evaluate the detection accuracy of the model for all categories. AP and mAP are both obtained when the IOU threshold is set to 0.5. GFLOPs indicates

TABLE 1. The AP comparison of the bounding circle model and the rectangular bounding box model on the NWPU VHR-10 dataset(%).

Methods	CenterNet	ACRFNet(w/o -FPN,-AC)
Airplane	98.82	98.80
Baseball field	94.91	98.22
Vehicle	90.22	91.34
Tennis court	85.58	81.96
Storage tank	79.97	85.65

the number of floating-point operations required for a single inference by the model, used to measure the computational complexity. Params represents the number of trainable parameters, usually measured in millions. Typically, the lower the GFLOPs and Params, the faster the detection speed.

C. VALIDATION FOR BOUNDING CIRCLE

Experiments are conducted on the NWPU VHR-10 and DIOR datasets to verify the effectiveness of the bounding circle in ACRFNet. Several circular and square-like targets are used for experimental comparison. These targets include airplane, baseball field, vehicle, tennis court, and storage tank. The results are shown in Table. 1. ‘-FPN’ indicates feature pyramid. ‘-AC’ indicates adaptive circular receptive field. ‘w/o’ is applied to indicate without these modules.

After using the bounding circle, the detection accuracy of the airplane target did not change significantly. Although the accuracy of the square and densely arranged tennis court target slightly decreased, the detection accuracy of the small square-like vehicle, scalloped baseball field, and circular storage tank has been improved. Among them, the storage tanks show the greatest improvement in detection accuracy, with an increase of 5.68 %.

The above experimental results indicate that using the bounding circle is more beneficial for detecting circular targets. Although vehicle targets are small squares in remote sensing images, bounding circles fit such targets well. For airplane targets, both rectangular and circular bounding box can represent them effectively.

The NWPU VHR-10 dataset has a small number of images. The number of difficult samples is also relatively small, which cannot effectively reflect the differences in model

performance. In order to further verify the effectiveness of the bounding circle, comparison experiments are conducted on the larger DIOR dataset for five types of targets: airplane, baseball field, vehicle, tennis court, and storage tank. The results are shown in Table. 2.

TABLE 2. The AP comparison of the bounding circle model and the rectangular bounding box model on the DIOR dataset(%).

Methods	CenterNet	ACRFNet(w/o -FPN,-AC)
Airplane	88.94	89.27
Baseball field	94.13	94.25
Vehicle	46.52	49.31
Tennis court	91.23	91.49
Storage tank	67.80	68.72

After using the bounding circle, the detection accuracy of the five types of targets has improved to some extent. The accuracy for the airplane increased by 0.33%. The accuracy for baseball field increased by 0.12%. The accuracy for the tennis court increased by 0.26%. The improvements for vehicle and circular target the storage tank are more significant, with the detection accuracy for the vehicle increasing by 2.79% and the detection accuracy for storage tank increasing by 0.92%.

The above experimental results indicate that the bounding circle can fit circular targets perfectly and suppress background interference effectively. The bounding circle is more suitable for detecting circular targets than the rectangular bounding box. Additionally, the bounding circle also exhibits good universality for square-like targets.

D. ABLATION EXPERIMENT

Since the adaptive circular receptive field and the feature pyramid are designed based on the bounding circle, and the adaptive circular receptive field is used in the feature pyramid, the ablation experiments are shown in Table. 3 and Table. 4. Ablation experiments are conducted on the NWPU VHR-10 and DIOR datasets. ‘BC’ indicates bounding circle. ‘FPN’ indicates feature pyramid. ‘AC’ indicates adaptive circular receptive field.

The ablation experiment results on the NWPU VHR-10 dataset are shown in Table. 3. The experimental results demonstrate that after adding the feature pyramid, the accuracy of detecting airplanes and baseball fields has increased to some extent, while the detection accuracy of tennis courts and storage tanks has significantly improved. Among them, the accuracy for the storage tanks improved by 8.96%. This indicates that the scale information brought by the feature pyramid is beneficial for the detection of targets such as storage tanks and tennis courts. On this basis, introducing an adaptive circular receptive field improved the detection accuracy of all five target categories. The accuracy of vehicle and storage tank detection has significantly improved. The vehicle increased by 5.66% and the storage tank increased by 3.04%. The above experimental results indicate that the adaptive circular receptive field is beneficial for feature

extraction of circular targets and has good compatibility with square-like targets.

The results of the ablation experiments conducted on the DIOR dataset are shown in Table. 4. After adding a feature pyramid to the bounding circle target detection model, the detection accuracy of airplanes, baseball fields, and tennis courts has improved. Vehicles are characterized by their smaller scale, and storage tanks are characterized by extensive scale variations. Therefore, the detection accuracy of these two types of targets has exhibited a notable improvement, with a 3.39% increase for vehicles and a 4.25% increase for storage tanks. By applying the adaptive circular receptive field to the feature pyramid, the detection accuracy of airplanes, baseball fields and tennis courts has increased to some extent. The accuracy of vehicle and storage tank has improved significantly, with increases of 3.10% and 4.80%, respectively. The above experimental data further indicate that the multi-scale information of the feature pyramid and the adaptive circular receptive field can effectively enhance the ability of the model to detect circular and square-like targets.

E. COMPARISON WITH OTHER METHODS

To verify the effectiveness and rationality of ACRFNet, the NWPU VHR-10 and DIOR datasets are used for experimental comparison with other methods. Typical methods include Faster R-CNN, RetinaNet, CornerNet [25], and YOLOX [26]. Advanced methods include T-TRD-DA [27], GLFE-YOLOX [28], and Sw-LBPN [29]. The results of the YOLOX method are referenced [30]. The categories of targets used for the experiment include airplane, baseball field, vehicle, tennis court, and storage tank.

The detection results on the NWPU VHR-10 dataset are shown in Table. 5. Compared to other methods, ACRFNet has significantly improved detection accuracy for circular targets and has also achieved good detection accuracy for other types of square-like targets. ACRFNet is generally superior to several other advanced methods.

Due to the relatively small number of samples in the NWPU VHR-10 dataset, there are relatively few images that show dense arrangements and large-scale variations. However, the DIOR dataset has a large number of samples, with more densely arranged and varying scale samples, which can better showcase the performance of the model. The experimental results on the DIOR dataset are shown in Table. 6. ACRFNet achieved the highest accuracy for the airplane, baseball field, tennis court, and storage tank targets, except for the vehicle target not achieving the highest accuracy. This further indicates that the bounding circle can better fit circular and square-like targets and carry less background information. The feature pyramid based on the adaptive circular receptive field enriches the circular target features and scale information in the feature map.

ACRFNet not only achieves high detection accuracy for specific circular targets but also has good compatibility

TABLE 3. Ablation experiments on the NWPU VHR-10 dataset(%).

BC	FPN	AC	Airplane	Baseball field	Vehicle	Tennis court	Storage tank
✓			98.80	98.22	91.34	81.96	85.65
✓	✓		99.02	97.96	88.06	89.74	94.61
✓	✓	✓	99.98	99.26	93.72	91.60	97.65

TABLE 4. Ablation experiments on the DIOR dataset(%).

BC	FPN	AC	Airplane	Baseball field	Vehicle	Tennis court	Storage tank
✓			89.27	94.25	49.31	91.49	68.72
✓	✓		91.45	95.47	52.70	92.73	72.97
✓	✓	✓	92.48	95.73	55.80	92.21	77.77

TABLE 5. Comparison with other methods on the NWPU VHR-10 dataset(%).

Methods	Airplane	Baseball field	Vehicle	Tennis court	Storage tank
Faster R-CNN	86.5	94.9	57.9	83.2	91.0
RetinaNet	89.5	99.0	66.1	78.1	85.7
CornerNet	81.6	95.2	75.2	78.4	70.3
CenterNet	98.8	94.9	90.2	85.6	80.0
YOLOX	99.4	95.8	88.9	94.8	92.9
T-TRD-DA	99.0	98.1	89.6	89.2	79.6
GLFE-YOLOX	99.9	81.7	84.3	96.7	97.5
Sw-LBPN	99.5	97.8	85.9	92.1	95.2
ACRFNet	99.9	99.3	93.7	91.6	97.6

TABLE 6. Comparison with other methods on the DIOR dataset(%).

Methods	Airplane	Baseball field	Vehicle	Tennis court	Storage tank
Faster R-CNN	54.1	63.3	43.1	83.2	53.3
RetinaNet	53.7	69.0	45.1	78.1	44.8
CornerNet	58.8	72.0	43.0	78.4	45.2
CenterNet	88.9	94.1	46.5	91.2	67.8
YOLOX	86.7	74.0	52.0	88.6	71.7
T-TRD-DA	77.9	70.1	50.5	84.0	69.8
GLFE-YOLOX	89.8	86.3	61.0	92.0	77.2
Sw-LBPN	82.8	75.0	56.9	87.3	75.9
ACRFNet	92.5	95.7	55.9	92.2	77.8

with square-like targets. In addition, the bounding circle avoids problems such as angular periodicity faced in rotating object detection in remote sensing images due to the rotational invariance of the circle. Table. 7 shows the overall accuracy results, the values of floating-point operations, and the number of parameters of ACRFNet compared to other methods on the NWPU VHR-10 and DIOR datasets. It can be observed that ACRFNet consistently achieves high accuracy in detecting all categories within the dataset. Benefiting from the concise and efficient CenterNet baseline, the GFLOPs and Params of ACRFNet are not very high. The GFLOPs of ACRFNet only increased by 1.1 compared to CenterNet. Since the prediction parameters of the detection head changed from width and height to just radius, the amount of prediction parameters has therefore decreased. The adaptive circular receptive field achieves higher detection accuracy with minimal computational cost.

TABLE 7. The mAP(%), GFLOPs, and Params(M) comparison of ACRFNet and other methods on the DIOR and NWPU VHR-10 datasets.

Methods	NWPU VHR-10	DIOR	GFLOPs	Params(M)
Faster R-CNN	87.6	63.1	134.4	41.1
RetinaNet	89.2	65.7	128.7	36.2
CornerNet	88.9	64.9	-	-
CenterNet	85.1	73.9	35.1	32.6
YOLOX	92.3	66.9	73.8	25.3
T-TRD-DA	87.9	66.8	125.7	-
GLFE-YOLOX	90.9	78.9	84.2	42.5
Sw-LBPN	93.8	73.9	117.0	45.1
ACRFNet	91.2	74.0	36.2	25.9

F. VISUALIZATION

Fig. 7 shows the detection results of CenterNet and ACRFNet on the NWPU VHR-10 dataset. The first row, second row, and third row show the visualization results for the baseball field, storage tank, and airplane, respectively. From the first

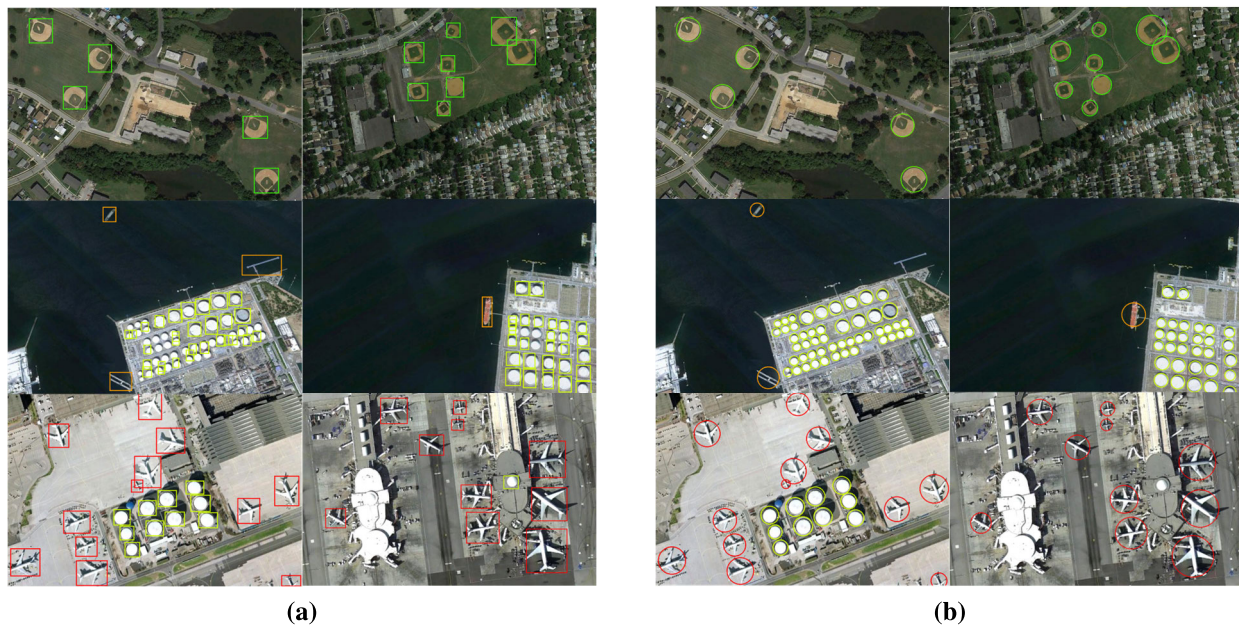


FIGURE 7. Comparison of visualization results between (a) CenterNet; (b) ACRFNet.

row, it can be observed that the rectangular bounding box of CenterNet does not fit the targets well and includes a lot of background information. The bounding circle of ACRFNet can better encompass the target, resulting in more precise localization. The second row shows the detection results of the storage tank targets. The storage tank in these two images varies in scale, is densely arranged, and there are many small storage tanks. The rectangular bounding box have many missed detections and inaccurate placements. In contrast, ACRFNet can fit the targets well and cope with changes in target scale and dense arrangement. The number of missed detections has been greatly reduced, and the positioning has become more accurate. The last row mainly shows the visualization results of the airplane. It can be seen that for targets with an aspect ratio close to 1, ACRFNet can also achieve accurate localization and good detection results.

Fig. 8 is used to further verify the detection performance of ACRFNet on densely arranged and square-like targets. The figure shows the detection results of vehicles and tennis courts using CenterNet and ACRFNet. The detection performance of CenterNet is quite unsatisfactory when detecting densely arranged vehicles with varying angles. In contrast, ACRFNet can achieve precise localization when faced with dense vehicle targets and can also detect vehicles against dark backgrounds. For rectangular tennis court targets, ACRFNet also demonstrates excellent compatibility.

To more intuitively demonstrate the advantages of ACRFNet in detecting densely packed circular targets, three satellite remote sensing images of nuclear wastewater storage tanks at a nuclear power plant are used for testing. Fig. 9 shows the results of detecting nuclear wastewater storage

tanks using CenterNet and ACRFNet. It can be seen that when faced with densely arranged nuclear wastewater storage tanks, the horizontal rectangular bounding box of CenterNet performs quite poorly. The localization is not accurate, with a significant number of missed detections and a few false detections. In contrast, the localization of ACRFNet is clearly more precise and does not have any false detections.

IV. DISCUSSION

Generalized remote sensing target detection methods have problems such as imprecise localization, difficulty adapting to scale variations, inclusion of excessive background information, and rotation angles in detecting circular and square-like targets. We propose ACRFNet for these types of problems. First, we propose the bounding circle instead of the rectangular bounding box to reduce background information and avoid rotation angle. Second, we propose the convolution of an adaptive circular sense field and add a feature pyramid to enhance the feature representation of circular and square-like targets. Finally, in order to rationally evaluate the model performance, we propose Circle-IOU to compute the mAP of the model. Experimental results on the NWPU VHR-10 with the DIOR dataset demonstrate the effectiveness of ACRFNet. The specific results are expressed as follows:

- For the airplane, baseball field, tennis court, vehicle, and storage tank, the bounding circle provides better detection accuracy than the rectangular bounding box. Among them, the circular storage tank shows the most significant improvement, increasing 5.68% at NWPU VHR-10 and 0.92% at DIOR.
- The results of the ablation experiments for five circular and square-like targets demonstrate the effectiveness

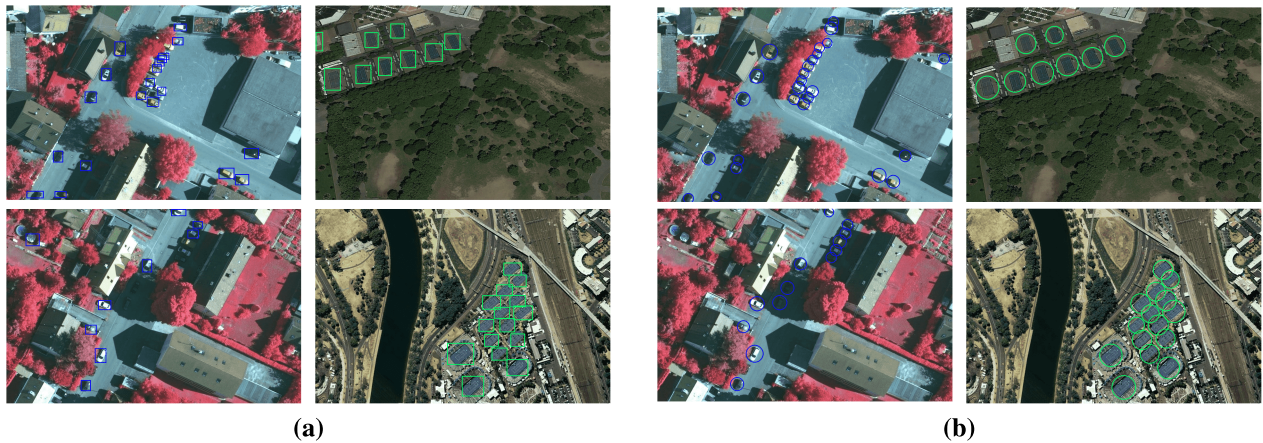


FIGURE 8. Dense vehicles and square tennis courts visualization detection results. (a) CenterNet; (b) ACRFNet.



FIGURE 9. Comparison of visualization results of nuclear sewage storage tanks between CenterNet(Top) and ACRFNet(Down).

of the feature pyramid. Except for the vehicle target in the NWPU VHR-10, which has a slight decrease in accuracy, the accuracy of all other targets is improved. After using the adaptive circular receptive field, the accuracy of five types of targets is improved.

- Comparative experimental results with other methods demonstrate that ACRFNet has optimal detection accuracy. ACRFNet achieves a high level of detection accuracy for all categories in the NWPU VHR-10 and DIOR datasets.
- The visualization demonstrates that ACRFNet can effectively handle situations with dense arrangement and scale changes, with more precise location and fewer missed detections. Moreover, the images of nuclear wastewater storage tanks are outside the dataset. The detection results demonstrate that ACRFNet has excellent generalization ability.

Our outlook for this work is as follows: Firstly, although ACRFNet has excellent detection accuracy, it sacrifices some of the detection speed and computational resources. The model size and detection speed are also very important. While improving the accuracy of the model, lightweighting the model is an important direction for future work. Secondly, in Fig. 9, ACRFNet has some missed detections due to the intraclass diversity of storage tanks. How to handle the similarity between different classes and the diversity between the same classes is one of the key factors in improving the robustness of the model. Data augmentation, attention mechanisms, and integrating contextual semantic information are all good choices for improving the generalization ability of the model. Data augmentation can increase the number of hard-to-classify samples. The attention mechanism can make the model pay more attention to the internal features of the target. Integrating contextual semantic information allows the model to adapt to complex scenarios. Finally, the bounding circle is

designed to better detect circular and square-like targets. For other targets with complex shapes, adaptive shape-changing detection boxes might be a good idea.

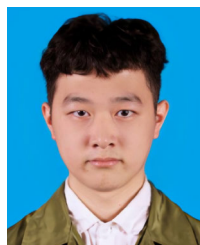
V. CONCLUSION

In remote sensing images, many pollution source buildings appear in circular shapes, such as storage tanks and nuclear wastewater storage tanks. These targets also have scale variations and dense arrangements. Therefore, a target detection method based on an adaptive circular receptive field, ACRFNet, has been proposed for such specific targets. First, the rectangular bounding box is transformed into the bounding circle. The bounding circle is better suited for fitting circular targets, suppressing background information interference, and avoiding rotation angles. Secondly, we propose the adaptive circular receptive field and combine it with the feature pyramid. This combination can not only better capture the features of circular targets but also effectively address scale variations and dense arrangements in remote sensing images. Therefore, the localization and classification of the model are more precise. Finally, Circle-IOU is proposed to calculate the IOU between two circular boxes. The performance of the model can be reasonably measured. Experimental results show that the proposed method achieved mAPs of 91.15% and 74.01% on the NWPU VHR-10 and DIOR remote sensing image datasets, respectively. The detection performance of circular targets has significantly improved. ACRFNet has achieved good detection performance on circular targets in remote sensing images and has good universality for overall category targets.

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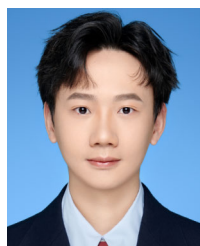
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